
Proposed Solution

1. Concept Summary

Opti-Crop is a web-based intelligence platform designed to replace agricultural guesswork with machine learning accuracy. Users input localized soil nutrient levels (N, P, K) and environmental metrics (pH, rainfall, temperature, humidity), and the system immediately generates a customized crop recommendation alongside detailed soil-health adjustments.

2. Key App Modules

Module A: Soil Predictor (Logistic Regression Classifier)

A page where users enter soil and climate conditions. The app queries a trained machine learning model and returns the crop most likely to achieve maximum yield, with customized watering and fertilizer guidelines.

Module B: Suitability Matrix

Allows farmers to "test-run" a crop of their choice. By entering soil values and selecting a target crop, the app displays a percentage compatibility score, indicating which parameters are optimal and which are deficient.

Module C: Cluster Insights Explorer

Provides research-oriented insights by grouping crops into 4 clusters based on soil parameters, showcasing which crops share similar environmental needs.

Module D: User Dashboard & Historical Logs

Provides persistent storage of prediction records, allowing researchers and farmers to trace soil parameters over time.